JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 85, NO. B7, PAGES 3681–3689, JULY 10, 1980

Ref 1228

Paleomagnetic Evidence for Tectonic Rotation of Jurassic Plutons in Blue Mountains, Eastern Oregon

DOUGLAS WILSON[1] AND ALLAN COX

Department of Geophysics, Stanford University, Stanford, California 94305

To test the hypothesis that pre-Tertiary rocks of eastern Oregon have undergone clockwise rotation, a paleomagnetic investigation was made of granitic and high-temperature metamorphic rocks of the upper Jurassic–Lower Cretaceous intrusive suite of the Blue Mountains province, northeastern Oregon. At nine sampling sites in three widely separated regions the paleomagnetic directions showed a high level of intrasite and intersite consistency, with a mean direction of $I = 63°$, $D = 30°$, $a_{95} = 10°$. The observed inclination is consistent with that expected on the basis of paleomagnetic results for stable North America, indicating that the intrusives have not undergone northward translation since cooling. The paleomagnetic declination, on the other hand, indicates that the region sampled has undergone a clockwise rotation of 60° ± 29° relative to the stable craton. Two of the bodies sampled are intrusive into the exotic Seven Devils terrane, and the remaining body is intrusive into the tectonic melange of a separate arc-trench terrane, which may well have been the North American margin. The consistency in magnetic directions between the three regions ($k = 151$) indicates that suturing of the exotic terrane took place prior to the emplacement of the intrusives. The early Tertiary Clarno Formation situated to the west of the region sampled has rotated substantially less than the Blue Mountains [*Beck et al.*, 1978], indicating that much of the rotation of the Blue Mountains occurred prior to the Eocene. To account for the observed rotations, we discuss two alternative models. In the first, rotation accompanies back arc rifting associated with the Early Cretaceous westward rifting of the Klamath Mountains. In the second, rotation reflects Late Cretaceous regional dextral shearing in the Blue Mountain area, which is located several hundred kilometers south of major right lateral faults known to extend from Alaska down at least to northern Washington.

INTRODUCTION

The Columbia Embayment is a triangular patch of Cenozoic volcanics and sediments comprising much of western Oregon and Washington (Figure 1). It is bounded on the north by the Mesozoic rocks of Washington and British Columbia, on the south by the Mesozoic rocks of the Klamath Mountains, and on the east by the Mesozoic rocks of the Blue Mountains of eastern Oregon. In his original paper defining oroclines, *Carey* [1958] presented a map showing the tectonic grain of the older terranes apparently wrapping around the margin of what we now call the Columbia Embayment, and he proposed that this area had undergone significant oroclinal bending. To test this hypothesis, *Watkins* [1965] sampled 237 lava flows of the Columbia Basalt at 11 widely separated sites in eastern Oregon and Washington. The orocline hypothesis predicts that the more southerly sites have undergone a clockwise rotation, the more northerly sites have not. *Watkins and Baksi* [1974] found that the ages of the basalts ranged from 16 to 13 m.y. and concluded that the paleomagnetic data were consistent with 15° clockwise rotation of the southern sites and no rotation of the northern sites. This would indicate a small amount of oroclinal bending during the past 13 m.y. However, *Watkins* [1965] was aware that a rotation this small was difficult to detect in the presence of geomagnetic secular variation and his final conclusion [*Watkins and Baksi,* 1974] was that the question of oroclinal rotation in eastern Oregon remains open.

The 15° of rotation reported by *Watkins* [1965] is considerably less than that suggested by the nearly east-west trend of the older Mesozoic terranes of the Blue Mountains in eastern Oregon. Therefore substantially larger rotations should be recorded paleomagnetically in the Mesozoic rocks if the orocline hypothesis or some variant of it is correct. The amount of rotation shown in the original map of *Carey* [1958] is about 60°. *Hamilton* [1969, 1978], and *Dickinson and Thayer* [1978] suggested a similar amount of rotation for the Blue Mountains. In Carey's model the rotation of the Blue Mountains was part of a much larger regional oroclinal bending, whereas in the models of Hamilton, Dickinson, and Thayer the rotation of the Blue Mountains is a more local tectonic phenomenon.

The present investigation was undertaken to determine whether the large amount of rotation suggested by the earlier studies is recorded in upper Mesozoic granitic intrusives and associated metamorphic rocks of the Blue Mountains.

GEOLOGIC AND TECTONIC SETTING

Older Terranes

The oldest rocks in the area are early Mesozoic volcanics and sediments that occur as two distinct terranes comprising two east-west trending belts (Figure 2). The southern terrane includes volcanics, clastics, and tectonic melange and is interpretted by *Dickinson* [1979] as an arc-trench assemblage produced by subduction of an oceanic slab toward the southeast during the Triassic and Early Jurassic. Whether this terrane formed along the edge of the continent or at some distance from it is uncertain [*Dickinson and Thayer,* 1978]. The northern belt, the Seven Devils terrane, consists mainly of Upper Triassic basalts and limestone and is thought by *Jones et al.* [1978] to be a possible southern extension of the exotoc 'Wrangellia' terrane that formed originally in low latitudes and presently comprises a belt extending from Vancouver Island to southeastern Alaska. Presumably, the subduction responsible for the formation of the southern belt is also associated with the transport of the northern Seven Devils terrane to this area.

[1] Present address: Institute of Geophysics, University of Hawaii, Honolulu, Hawaii 96822.

Copyright © 1980 by the American Geophysical Union.

Paper number 9B1694.
0148-0227/80/009B-1694$01.00

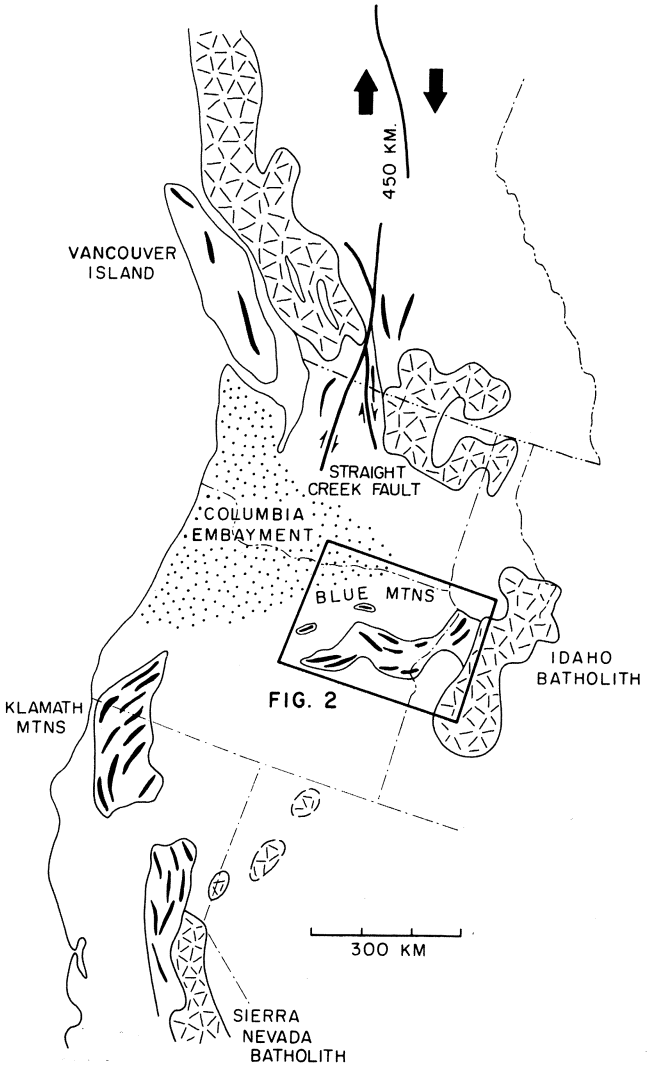


Fig. 1. Tectonic setting of the Blue Mountains. Columbia Embayment is area underlain by Cenozoic volcanics and sediments. Tectonic trends are marked by solid lenticular pattern in the Blue Mountains and elsewhere.

Intrusives

Both the southern and northern terranes are intruded by the Upper Jurassic and Lower Cretaceous granitic intrusives which are the subject of the present study. These intrusives indicate that by the Late Jurassic, subduction apparently had shifted to the west. Unless the intrusions into the different terranes are unrelated, it appears likely that the older terranes were assembled by the Late Jurassic. One of the objectives of the paleomagnetic investigation was to test this interpretation: If the Seven Devils terrane had been intruded prior to its coming to rest in eastern Oregon, then discordant paleomagnetic directions would be expected from the intrusions in the northern and southern terranes.

The intrusions of the Blue Mountains [*Taubeneck,* 1957, 1964*a*, *b*] consist largely of granodiorite, tonalite, and trondhjemite and include the composite Bald Mountain and Wallowa batholiths and numerous smaller bodies. Potassium-argon and rubidium-strontium ages range mainly from 160 to 125 m.y., although a few K-Ar ages are as young as 93 m.y. [*Armstrong et al.,* 1977]. Many of the younger ages are discordant with field relations establishing relative ages of intrusion. It appears likely that the younger ages do not record the age of initial intrusion but rather reflect the effects of extremely slow cooling, reheating during protracted magma emplacement, hydrothermal alteration, or partial resetting during Cenozoic volcanism.

Younger Rocks

The youngest exposed Mesozoic rocks in the Blue Mountains are shallow water-laid mid-Cretaceous sediments preserved in various localities in the western Blue Mountains. The most extensive deposits occur near Mitchell (Figure 2), where a section 2.7 km thick was deposited in a rapidly subsiding basin in mid-Cretaceous time [*Wilkinson and Oles,* 1968]. The oldest of the Cenozoic volcanics in this area is the Eocene Clarno Formation, a largely andesitic suite exposed west and north of Mitchell. Most of northern Oregon is blanketed by tholeiitic flood basalts of the Miocene Columbia River Basalt, which reaches a thickness of several kilometers in northern Oregon and southern Washington and terminates to the south in the Blue Mountains.

Tectonic Corrections

Before paleomagnetic results can be rigorously interpreted in terms of oroclinal rotations, it is necessary to determine whether the plutons have undergone large tilts about horizontal axes. At one locality the Miocene Columbia Basalt laps onto the pluton and is observed to dip about 5°, providing a minimum tectonic correction. However, generally there is no direct tectonic information in the form of stratiform beds slightly younger than the intrusions deposited directly on the plutons.

The available indirect structural data suggest that deformation was not very intense at our sampling sites. The Miocene formations in this area are deformed by broad, gentle folds with NE to ESE trends and have undergone block faulting on NW trending normal faults [*Walker,* 1977; *Brown and Thayer,* 1966]. The sampling localities are at or near the cores of these gentle folds or in areas of broad uplift, so tilting since the Miocene has probably been small. It is noteworthy that at the time of extrusion of the Columbia River Basalt the Blue Mountains already existed as a topographic high, as indicated by the onlap of the Columbia Basalt [*Waters,* 1962]. Also, in the western Blue Mountains, present structural and topographic highs have persisted since at least the Oligocene and probably since the Eocene [*Fisher,* 1967], suggesting tectonic stability during at least the later history of this region. Less is known about the earlier tectonic history of this area, and in our final interpretation we will assume that the amount of tilting was small, as is commonly done in interpreting paleomagnetic data from intrusives. Our paleomagnetic data will provide a basis for testing this assumption. If it is incorrect and a large amount of tilting has, in fact, occurred, it seems likely that the direction and amount of tilting would vary over an area this large, so that the different plutons would be expected to have significantly different directions of magnetization.

THIS STUDY

During the summers of 1977 and 1978, six plutonic units and their contact aureoles were sampled at 27 sites, all in road cuts or glaciated mountains where weathering was minimal. Six core samples were generally drilled at sites in the intrusives, and four block samples were generally collected at sites in the metamorphics. The precision of orientation is estimated

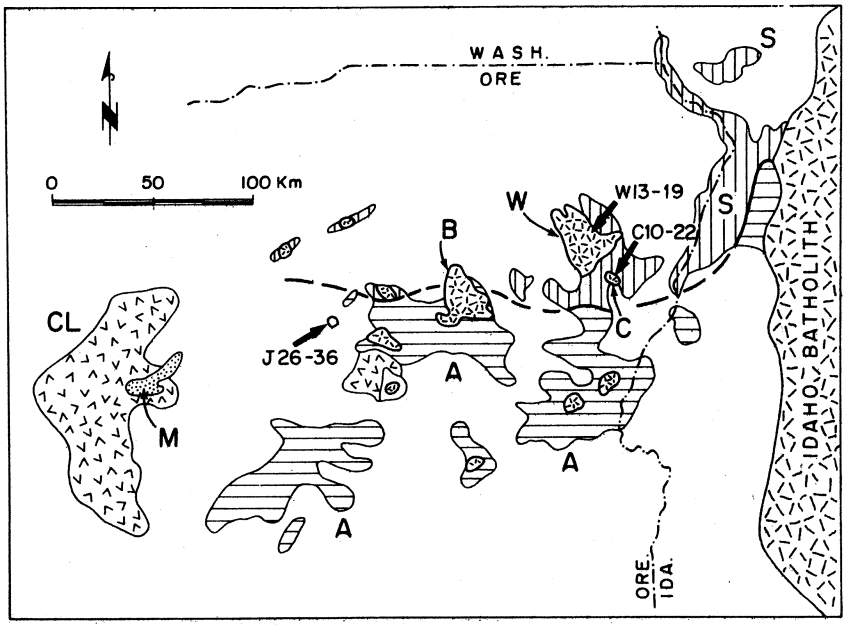


Fig. 2. Regions sampled in the Blue Mountains, W, Wallowa batholith (2 sites); C, Cornucopia stock (4 sites); J. western sampling area (3 sites); S, Seven Devils terrane; A, southern arc-trench assemblage; M, Cretaceous sediments near Mitchell; and CL, Clarno Formation.

to be 2° for the core samples and 5° for the blocks. Individual specimens 25 mm in diameter and 24 mm long were prepared from both types of samples.

Measurements of remanent magnetization were made on a two-component cryogenic magnetometer. Magnetic cleaning in peak fields up to 100 mT was done using demagnetization coils built into the magnetometer. At each field level the specimen was demagnetized successively along three orthogonal axes and the remanence then measured. Sometimes the demagnetization was repeated to check for consistency. Thermal demagnetization was done in air in an ambient field of 15 nT or less, holding the specimen at the maximum temperature during each run for 30 min.

Unstable and Inconsistent Sites

At some of the sites it was impossible to obtain consistent paleomagnetic directions for one of two reasons. (1) Although the magnetic directions of individual specimens were extremely stable on partial demagnetization to high fields and high temperatures, the magnetic directions within a site would never converge, the intrasite scatter in directions typically being 50°–100°. At one site, changes in direction during thermal demagnetization suggested that the observed dispersion in direction may have been the result of having recorded a transitional direction of the field between reversals. For the other unstable sites, including all of the sites from the Bald Mountain batholith, some of the sites in the Wallowa batholith, all of the sites in the contact aureole of the Wallowa batholith, and several sites in smaller unnamed intrusions, we are at a loss to explain the stable, highly scattered magnetic directions that we observed. (2) Specimens from several sites did not yield consistent magnetic directions after repeated partial demagnetization at the same peak field, indicating that no observable stable component of magnetization was present. Eighteen of the total of 27 sites did not yield internally consistent paleomagnetic data.

Stable and Consistent Sites

At the remaining nine sites, metamorphic and granitic rocks and mafic inclusions were found to be magnetically stable and to display intrasite consistency in their magnetic directions. Thermal demagnetization experiments on specimens of granitic and mafic rock usually displayed blocking temperatures somewhat below 580°C, indicating that magnetite is the magnetic carrier (Figure 3). In one of the metamorphic specimens, blocking temperatures above 580°C were encountered (Figure 3), indicating that at least part of the remanence is carried by hematite. Good consistency in magnetic directions was noted between the remanence carried by magnetite and hematite. The magnetic coercivity of remanence, as measured by the decay of remanence in alternating fields, was generally higher in mafic inclusions than in granodiorite (Figure 3), but all three rock types were found to yield internally consistent data.

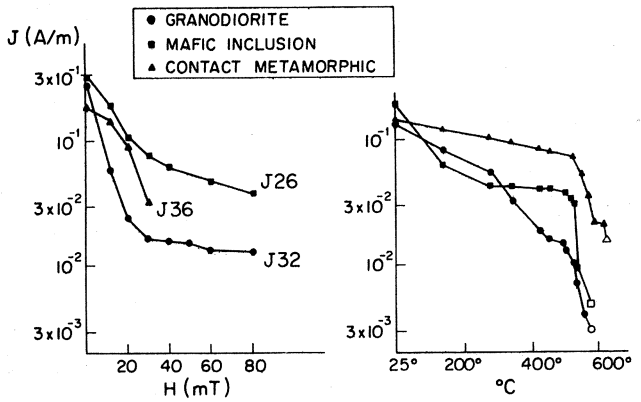


Fig. 3. Intensity J of magnetization of typical specimens after partial demagnetization in alternating fields (peak fields shown in milliTeslas) and after partial thermal demagnetization. Open symbols indicate erratic magnetizations discordant with characteristic directions produced by experimental noise at high fields and temperatures.

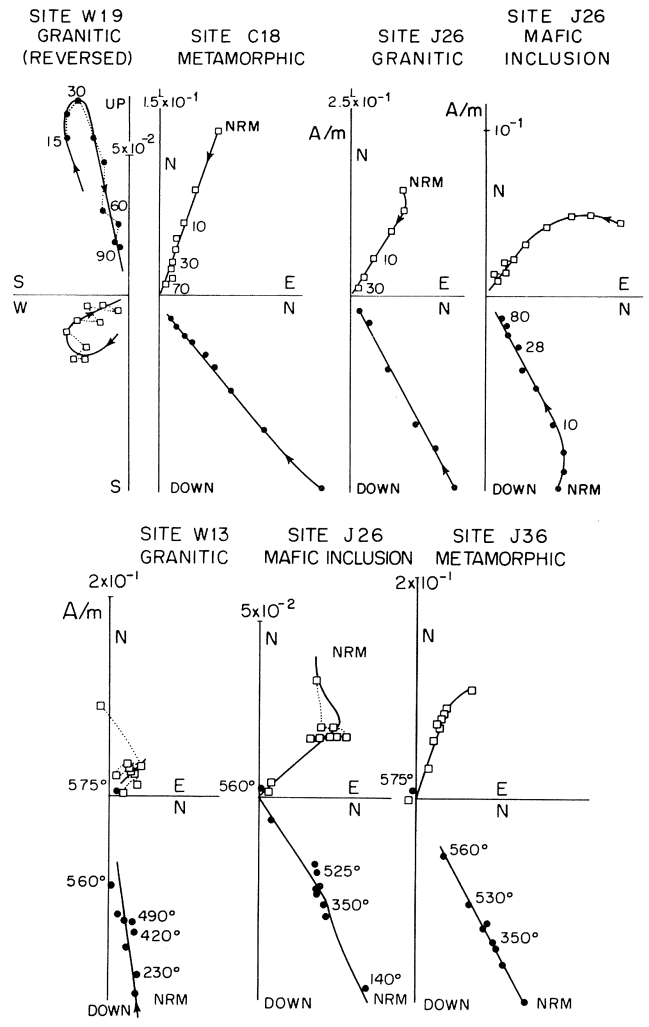


Fig. 4. Zijderveld component diagrams of partial demagnetization experiments, (top) peak alternating fields shown in milliTeslas and (bottom) temperatures shown in degrees Centrigrade.

To determine the stable direction of magnetization for each site, Zijderveld component plots were used to identify the characteristic direction of each specimen [*Zijderveld,* 1967]. Pilot specimens from each group of sites were first partially demagnetized at closely spaced peak alternating fields and temperatures. At the stable sites, thermal and alternating field demagnetization were used on several different specimens, and both generally gave essentially the same characteristic directions. On the basis of these experiments the remaining specimens were demagnetized stepwise until the directions of magnetization became erratic, which generally occurred only after the intensity of magnetization had been reduced by a factor of 5 or more. The intensity of several specimens decreased by less than a factor of 2 on demagnetization to 100 mT, and data from these were included in the analysis. Zijderveld plots were then used to identify characteristic directions if the data were coherent (Figure 4). At a few sites the plots revealed that an overprint of a single unstable component had been removed, but otherwise the magnetic interpretation was straightforward. Directions judged to represent the characteristic direction of each specimen were averaged to determine a single direction for the specimen. The range of demagnetization levels used to determine the characteristic direction is listed in the appendix for each specimen along with descriptions of the individual sampling sites.

Polarity

One of the larger plutons sampled appears to have cooled slowly enough to have recorded both normal and reversed polarity. Both reversely magnetized site W19 and normally magnetized site W13 are from the Hurricane Divide pluton of the Wallowa batholith and are situated only 1 km apart.

Both normal and reversed magnetizations were found in single specimens from site C18 in the contact aureole of the Cornucopia stock, suggesting that the geomagnetic field reversed during the cooling of these rocks. During progressive heating, specimen C21B from this site switched polarity several times, indicating that both normal and reversed components of magnetization were present in different blocking temperature ranges (Figure 5). The normal components of magnetization have higher intensities of magnetization and are better grouped in direction, so normal magnetization is clearly dominant. Another specimen from the same site C19B1, although retaining normal magnetization during step heating, showed an increase in magnetization for the three steps over the temperature range 420°–500°C, indicating the presence of reversed magnetization. Apparently, the geomagnetic field changed polarity at least once as this site cooled but was normal during the acquisition of most of the remanence.

The presence of both normal and reversed polarity with a strong bias toward normal polarity helps constrain the age of these intrusions, which are rather loosely dated radiometrically between 160 and 125 m.y. before present. According to the Jurassic reversal time scale of *Van Hinte* [1976] the field was normal all of the time from 164 to 150 m.y. before present, indicating that the time when the intrusives cooled through their magnetic blocking temperatures, approximately 500°–570°C, was less than 150 m.y. B.P. The time required for the cooling and magnetization of all of the sampled sites is highly uncertain. It is probably more than 1 million years and more likely is of the order of 10 million years. Viewing the Late Jurassic and Early Cretaceous time scale through a sliding window 10–20 m.y. long, the interval that best satisfies the condition that the field was reversing but of dominantly normal polarity starts at the beginning of anomaly M22 of *Larson and Hilde* [1975] and terminates at the end of anomaly M14. According to the time scale of *Larson and Hilde* [1975] the corresponding age range is 148–132 m.y., and according to that of *Van Hinte* [1976] the range is 142 to less than 132 m.y. A mean age of cooling of these intrusions of 140–130 m.y. would thus be consistent with the magnetic polarity information.

Directions

The following procedure was used in reducing the paleomagnetic data to obtain a mean paleomagnetic direction for this region. After determining the characteristic direction for each specimen, the specimen directions were combined using Fisher statistics to obtain a magnetic direction for each oriented sample. These sample directions were then combined to obtain the site means listed in Table 1. Tectonic corrections were available for only two sites, W13 and W19, based on the attitude of the on-lapping Columbia River Basalt. This tectonic correction reduced slightly but not significantly the intersite scatter ($k = 92$ versus $k = 69$). Quite possibly the scatter would be further reduced if tectonic corrections were available at all sites. However, the partially corrected paleomagnetic directions are in very good agreement, indicating that the amount of differential tilting throughout the areas sam-

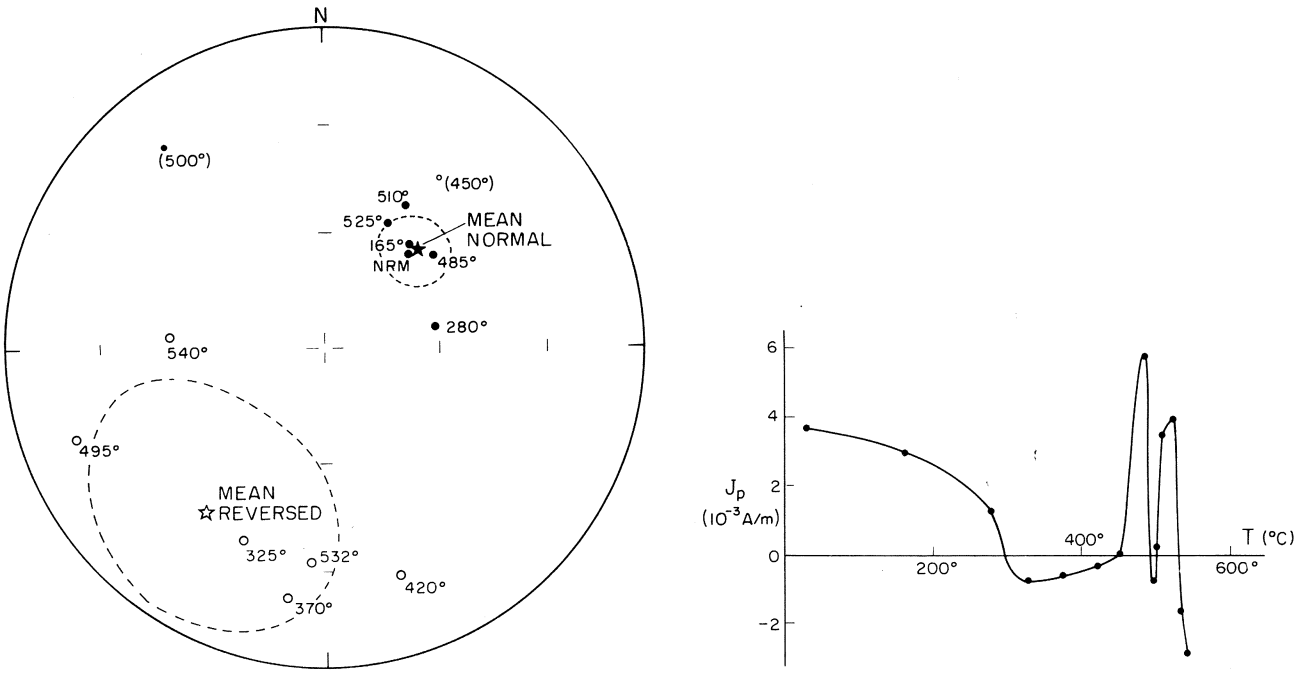


Fig. 5. Partial thermal demagnetization of specimen C21B. Normal components of magnetization (solid circles, positive inclinations) and reversed components (open circles, negative inclinations) are present in different ranges of blocking temperatures. J_p, the component of magnetization parallel to the normal site mean direction, changes sign 5 times. The more divergent magnetic vectors generally have lower intensities of magnetization. After heating to 450° and 500° (small circles), intensities were less than 5% of the maximum values, the normal and reversed components remaining at these temperatures being nearly balanced.

pled has not been large. Tilting is still probably the greatest source of experimental error, and in recognition of this we have, in our final statistical analysis, first combined the directions of all sites from a given region and then statistically combined the mean directions from the three regions. If the nine site means are treated as independent data, the resulting a_{95} confidence circle is 5°, whereas if the three regional means are analyzed, a_{95} increases to 10° (Table 1). The latter is probably a more realistic estimate of the error.

INTERPRETATION

Comparison With Stable North America

Inclination. To provide a basis for comparing these paleomagnetic results with those from stable North America, paleomagnetic poles for 130, 140, and 150 m.y. B.P. and their 95% confidence circles [*Irving,* 1979] were mapped into the expected field directions and confidence ovals at the sampling site. The seimiaxes of the oval about the expected field direction, a_1 in the direction of increasing and decreasing inclination and a_2 in the direction of increasing and decreasing declination, were found from A_{95}, the confidence circle about the North American pole, by the following equations. The uncertainty ΔI in inclination is simply

$$\Delta I = a_1 = 2A_{95}/(1 + 3\cos^2 p_0) \quad (1)$$

where the paleocolatitude p_0 is the distance from the sampling site to the expected paleopole. The uncertainty ΔD in declination is

$$\Delta D = \sin^{-1}(\sin A_{95}/\sin p_0) = \sin^{-1}(\sin a_2/\cos I_0) \quad (2)$$

where I_0, the expected inclination at the site, is given by

$$I_0 = \tan^{-1}(2\cot p_0) \qquad -90° \leq I_0 \leq 90° \quad (3)$$

Note that for steep inclinations, ΔD is much larger than a_2. The paleomagnetic inclination of the Blue Mountain intrusives is not significantly different from the inclination inferred from North American data (Table 1), indicating that the intrusives have not been transported a great distance toward or away from the paleomagnetic pole. This is consistent with *Dickinson*'s [1979] conclusion based on geologic analysis that the intrusives had formed in situ. However, the constraint provided by the paleomagnetic data is not a very strong one because of the large confidence circles of the north American and Blue Mountain poles, both of which are about 13°. The uncertainty in the amount of poleward displacement of the sampling area is equal to the root mean square (rms) sum of the errors in the two poles, which is $(13^2 + 13^2)^{1/2} = 18° = 2000$ km. The present data would not be able to resolve displacements smaller than this with 95% confidence.

Declination and rotation. The difference between the observed and calculated declinations indicates quite clearly that the Blue Mountains have undergone a statistically significant clockwise rotation. To assess the amount of rotation, we used the paleofield directions and confidence ovals at the sampling site for 150, 140, and 130 m.y. (Figure 6). The corresponding 95% confidence intervals for inclination and declination are listed as ΔI and ΔD in Table 1. If D_0 and ΔD_0 are the observed paleomagnetic declination and its uncertainty and if D_c and ΔD_c are the calculated declinations for stable North America, then the rotation angle R and its uncertainty ΔR are given by

$$R = D_0 - D_c \quad (4)$$

$$\Delta R = (\Delta D_0^2 + \Delta D_c^2)^{1/2} \quad (5)$$

Using the values of ΔD listed in Table 1, the resulting values of $R \pm \Delta R$ for 150, 140, and 130 m.y. are 50 ± 26, 58 ± 28,

TABLE 1. Summary of Results

	n	N	I	D	a_{95}	k
Site						
C10	7	4	53°	24°	10°	78
C14	7	3	64°	29°	12°	108
C18	8	4	56	34°	7°	160
C22	7	4	69	27°	15°	41
J26	11	6	56	44°	6°	108
J32	3	3	58	26°	10°	161
J36	11	6	60	11°	3°	406
W13	7	6	77	37°	2°	853
W13 (tectonically corrected)	7	6	73	27°	2°	853
W19	8	6	−70	235°	4°	272
W19 (tectonically corrected)	8	6	−67	225°	4°	272
Mean of sites						
Tectonically uncorrected		9	63	31	6	69
Tectonically corrected		9	62	30	5	92
Mean of regions						
Tectonically uncorrected		3	65	32	14	75
Tectonically corrected*		3	63*	30*	10*	151*
Expected Paleofield†			I	D	ΔI	ΔD
150 m.y.			58°	340°	10°	14°
140 m.y.			63°	332°	11°	18°
130 m.y.			65°	327°	10°	19°

Here n is the number of specimens measured, N is the number of field-oriented samples, a_{95} is the confidence circle, and k is the precision parameter.

*Best estimate of paleofield.

†Based on paleopoles by *Irving* [1979]. The ΔI and ΔD are 95% confidence intervals for I and D corresponding to A_{95} of poles listed by *Irving* [1979]. Transforming the three site means to poles, the mean pole ($N = 3$) has coordinates (69°N, 321°E) with $A_{95} = 13°$.

and 63 ± 29. Interpolating between the values for 140 and 130 m.y., the most likely age range of the intrusives on the basis of their magnetic polarity as previously discussed, the amount of clockwise rotation is found to be 60 ± 29, where the range indicates our estimate of the 95% confidence interval.

Time of Accretion of Early Mesozoic Terranes

The good agreement in paleomagnetic directions from the three regions sampled permits several conclusions to be drawn about the time of accretion of the northern exotic Seven Devils terrane and the southern arc-trench terrane. The C sites from the Cornucopia stock and the W sites from the Wallowa batholith are in rocks instrusive into the Seven Devils terrane, whereas the other sites are in rocks intrusive into the southern arc-trench terrane. The good agreement between the two sets of paleomagnetic directions suggests that very little tectonic displacement between the two terranes has taken place since the time of intrusion. The suturing of the Seven Devils terrane to the southern arc-trench terrane apparently took place prior to the interval 160–125 m.y., the age range of the intrusions. Although the question of whether the southern arc-trench terrane formed in its present location relative to North America is still open, the agreement of our paleomagnetic inclination with the stable North America data suggests that this terrane was roughly in its present location by the Late Jurassic, subject to the uncertainties discussed earlier.

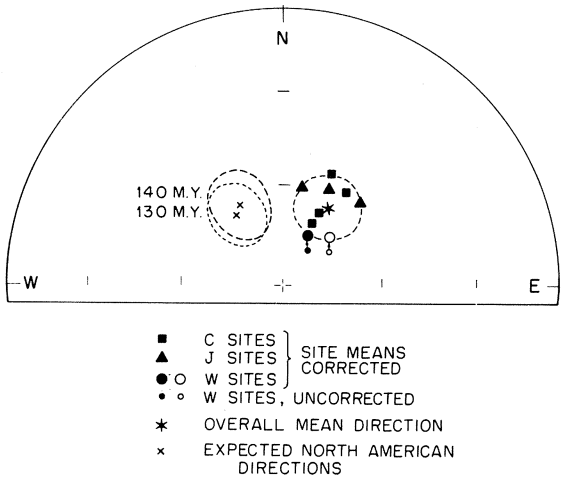


Fig. 6. Mean directions of magnetization for nine stably magnetized sites. Post middle Miocene tectonic corrections (small arrows) are known for the two W sites, one of which (open circle) is reversed and is reflected here onto the lower hemisphere for comparison with the normal sites. The geomagnetic field directions expected in eastern Oregon in the Jurassic on the basis of data from the stable North American craton are shown by crosses and dashed 95% confidence ovals.

Orocline Hypothesis

The difference between the observed and expected declinations can best be explained by a clockwise rotation of approximately 60° about a vertical axis since the end of the Jurassic. To produce the observed paleomagnetic direction entirely by rotation about a horizontal axis would require a 30° tilt toward the west about a north-south axis. The total area sampled is much too large to have tilted this much as a single block. The agreement between results from the three different regions and the absence of north-south tectonic trends of any age argue against the tilting or folding of smaller blocks by the large amount needed to account for the observed declinations. We conclude that the paleomagnetic rotation predicted by the orocline hypothesis is clearly present in the intrusives of the Blue Mountains. However, the mechanism by which the rotation occurred is still an open question, and this will be considered in the light of other geological and geophysical information, much of which was not available at the time of *Carey*'s [1958] paper.

TABLE 2. Middle Miocene Paleofields in Eastern Oregon

	Pole		Field Direction					
	North Latitude	East Longitude	a_{95}	p	I	D	ΔI	ΔD
Observed	83°	325°	7.4°	45°	64°	10°	5.9°	10.6°
Calculated	87°	167°	5.0°	44°	64°	−4°	3.9°	7.2°

The observed pole is the mean of paleomagnetic poles of sites SM, OR, PG, M, and I of *Watkins and Baksi* [1974]; the calculated pole is the Miocene (20 m.y.) pole of *Irving* [1979]; p is the colatitude or distance from pole to sampling site. Field directions are the following: I and D were found from the observed and calculated poles for the mean Blue Mountain site coordinates (45°N, 118°W). The mean direction of the five sites ($I = 63$, $D = 10$) is essentially the same as the 'observed direction' found from the pole for the Columbia basalts. $\Delta I = 2A_{95}/(1+ 3 \cos^2 p)$ and $\Delta D = \sin^{-1}(\sin A_{95}/\sin p)$.

Previous Paleomagnetic Investigations

Paleomagnetic research in the Cordillera has shown that clockwise rotations of crustal or lithospheric blocks are the rule rather than the exception [*Beck,* 1976]. Moreover, in a tectonically complex area like the Blue Mountains the net rotation observed paleomagnetically may be the result of several distinct tectonic events occurring at different times. The present result should therefore be considered in the context of tectonic rotations established by paleomagnetic studies in adjacent regions, which help establish the timing of the rotation and the size of the rotated crustal block. The following is a summary of the paleomagnetic results from nearby areas.

Miocene Columbia River Basalt. Five of the sites of *Watkins and Baksi* [1974] lie within or somewhat south of the Blue Mountains in eastern Oregon. The data from these sites (Table 2) yield a mean middle Miocene direction for eastern Oregon of $I = 64°$, $D = 10°$. The expected field direction in the middle Miocene based on paleomagnetic data from stable North America (Table 2) is $I = 64°$, $D = -4°$. The indicated clockwise rotation is thus 14°, in accord with the interpretation of *Watkins and Baksi* [1974], with an uncertainty at the 95% confidence level of ±13°, taking into account the uncertainty in declination of both the observed and calculated paleofield directions (Table 2).

To determine whether the rotation of the Blue Mountains is significantly greater than that of the Columbia River Basalt, we note that if $R_B \pm \Delta R_B$ is the rotation of the former and $R_C \pm \Delta R_C$ is that of the latter, the difference ${}_CR_B$ between the two rotations and the uncertainty in this difference at 95% confidence is given by

$$ {}_CR_B = (R_B - R_C) \pm (\Delta R_B{}^2 + \Delta R_C{}^2)^{1/2} = 46° \pm 32° $$

This would indicate that at least part of the rotation of the Blue Mountains, and possibly all of it, occurred prior to the middle Miocene.

Clarno Formation. The Clarno Formation (Figure 1) to the west of the Blue Mountains includes andesitic flows of Eocene age that have been studied paleomagnetically by *Beck et al.* [1978], who report clockwise rotation of 15° ± 22°. The difference between this and the Blue Mountain rotation is 45° ± 36°. Again the Blue Mountain rotation is greater than that in adjacent Tertiary rocks, indicating in the case of the Clarno that at least part of the Blue Mountain rotation is pre-Eocene.

Oregon Coast Range. Clockwise rotation since the Eocene of 50°–60° are well documented in the Oregon Coast Range [*Cox,* 1957; *Simpson and Cox,* 1977; *Beck and Burr,* 1979; *Cox and Magill,* 1977; *Plumley and Beck,* 1977]. Although the amount and sense of rotation is as great as in the Blue Mountains, the two rotations probably occurred at different times and in different tectonic settings, so it is problematical whether they are related.

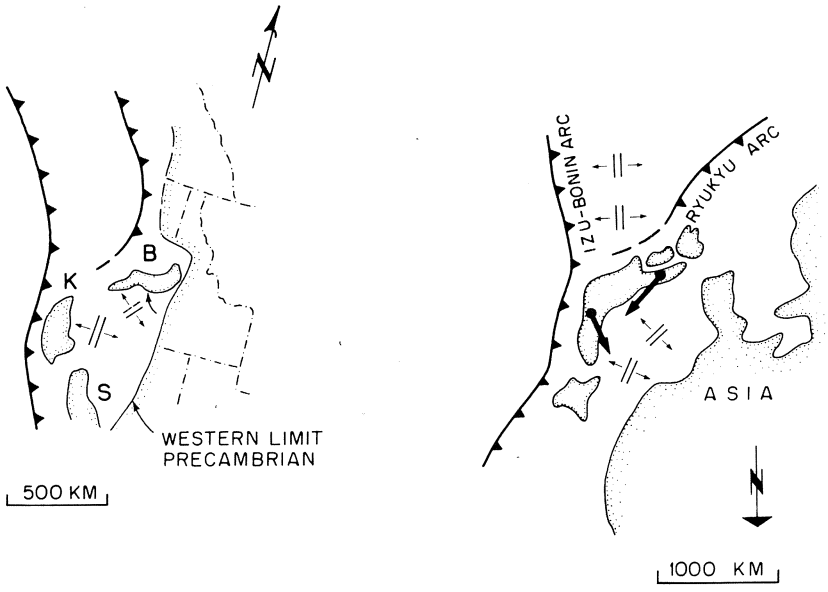


Fig. 7. (Left) Rotation of the Blue Mountains (B) may have accompanied Mesozoic rifting of the Klamath (K) and Sierra Nevada (S) Mountains away from the older craton. (Right) Analagous middle Tertiary rifting of Japan away from Asia, accompanied by differential rotation of northern and southern Honshu, are established by paleomagnetic results (arrows) from different parts of the island [*Kawai et al.,* 1969].

Regional Tectonics

To explain these data, we propose two models which involve significant rotation between the Late Jurassic and Eocene. The first is based largely on the paleogeographic reconstruction of *Hamilton* [1969, 1978] and *Dickinson and Thayer* [1978] in which the Blue Mountains pivot clockwise at their northeastern end in a rifting event that may or may not have been accompanied by westward motion of the Klamath Mountains (Figure 7). As originally proposed by *Hamilton* [1969], the rifting and westward motion of the Klamaths occurred in the late Cenozoic. However, *Blake and Jones* [1977] concluded from geologic field relations that the Klamaths had moved westward away from the Sierra Nevada before the Aptian (late Early Cretaceous). An interpretation of our data from the Blue Mountains consistent with the above is that the rifting of the Blue Mountains occurred in the Early Cretaceous, possibly in a back arc setting analogous to the opening of the Sea of Japan during the Cenozoic.

The paleogeography of the Pacific northwest during the Mesozoic is reminiscent of the present plate tectonics of the Japanese, Izu-Bonin, and Ryuku arcs (Figure 7). Spreading in the marginal sea between Japan and the continent not only rifted Japan away from the continent but, on the basis of paleomagnetic data [*Kawai et al.*, 1969], appears to have produced differential rotation between the northern and southern parts of the arc. The analogy is strengthened if, as *Davis et al.* [1978] have suggested on the basis of the patterns of sedimentation and magmatism in northern Washington and British Columbia, there were two parallel subduction zones during the Late Jurassic and Early Cretaceous.

A less specific alternative model relates the observed clockwise rotation to the general northward translation during the Mesozoic of slivers of accreted continental margin along a zone of dextral strike slip faults marking the western margin of North America. This system of faults with large dextral displacements extends from Alaska to northern Washington, where the Straight Creek Fault (Figure 1) terminates beneath the Tertiary rocks of the Columbia Embayment. *Davis et al.* [1978] suggested that in the latest Cretaceous the Straight Creek Fault extended from British Columbia southward to the vicinity of the Blue Mountains. Terrane that lay west of the Blue Mountains during the middle Mesozoic in what is now the Columbia Embayment may have moved northward toward the end of the Mesozoic. If this is correct, it is quite possible that the Blue Mountains, separated from the terrane to the west by a dextral fault, experienced clockwise rotation owing to regional dextral shear in a manner analagous to the well-documented Tertiary clockwise rotation of terrane adjacent to the San Andreas Fault in southern California [*Kamerling and Luyendyk*, 1979; *Greenhaus and Cox*, 1979].

Attempts to make detailed palinspastic reconstructions for the Mesozoic by sliding northwestern Washington and British Columbia southward along the Straight Creek Fault and adjacent faults (Figure 1) encounter a space problem: the northern terranes overlap the western part of the Blue Mountains [*Davis et al.*, 1978]. This overlap does not occur if in reconstructions for the Mesozoic the Blue Mountains are rotated back counterclockwise from their present position about a local Euler pole near the northeast end of the range. This requires that the rotation must have occurred after the northward motion of the terrane west of the dextral fault along the west edge of the Blue Mountains. However, the post-Eocene rotation inferred from paleomagnetic data from the Clarno Formation is probably adequate to permit self-consistent paleogeographic reconstructions, so resolving the overlap problem by rotating the Blue Mountains does not add much to out understanding of when the Blue Mountains rotated.

The reported absence of substantial rotation in the Eocene Clarno Formation, situated between the rotated Oregon Coast Range and the rotated Blue Mountains, points to models like the above in which the two rotations occurred at different times by different mechanisms. If on closer scrutiny the Clarno Formation turns out to have rotated also, it may be simpler to model all of the rotations in Oregon by a common tectonic process occurring during the Cenozoic.

APPENDIX

For each site the following information is listed in sequence: site number, site location, rock type, nature of outcrop, longest dimension of site, and the range of magnetic or thermal treatment levels used to determine the characteristic directions for each specimen from that site. For example, at site C14 the characteristic direction for the first specimen was determined from points on a component plot beginning with the natural remanence N and ending with the magnetization after heating to 485°C; for the third specimen, partial magnetizations were used after demagnetizations in fields ranging from 20- to 90-mT peak field.

Wallowa Batholith-Hurricane Divide
(Sites 1 km Apart)

Site W13 (45°15.1′N, 117°22.3′W). Tonalite. Specimens span 15 m on hillside outcrop. Treatment: 150°–550°C, 150–575°C, N–40 mT, N–50 mT, N–40 mT, N–50 mT, 5–60 mT.

Site W19 (45°15.5′N, 117°23.0′W). Tonalite. Specimens span 5 m in stream bank exposure. Treatment: 340°–550°C, 400°–550°C, 10–100 mT, 10–40 mT, 10–100 mT, 15–50 mT, 30–90 mT, 10–60 mT.

Western Sampling Area
(Sites 1 km Apart)

Site J26 (44°51.1′N, 119°2.7′W). Granodiorite with mafic inclusions. Specimens span 15 m in road cut. Treatment: 300°–580°C, 150°–563°C, 20–60 mT, N–30 mT, 3–30 mT, N–30 mT, 10–80 mT, 20–80 mT, 5–33 mT, 28–60 mT, N–28 mT.

Site J32 (44°50.7′N, 119°2.4′W). Granodiorite. Specimens span 10 m in road cut. Treatment: N–28 mT, 5–40 mT, 5–28 mT.

Site J36 (44°50.7′N, 119°2.5′W). Metamorphosed ultramafics 20 m from contact with pluton. Specimens span 5 m in road cut. Treatment: 150°–600°C, 150°–490°C, N–30 mT, 5–30 mT, N–20 mT, N–20 mT, N–20 mT, N–20 mT, N–20 mT, N–33 mT, N–30 mT.

Cornucopia Stock
(Sites Span 5 km)

Site C10 (45°2.9′N, 117°15.5′W). Schist 10–40 m from contact with pluton. Specimens span 30 m on hillside outcrop. Treatment: N–20 mT, 80–100 mT, 15–90 mT, 60–100 mT, 20–70 mT, N–90 mT, N–90 mT.

Site C14 (45°2.9′N, 117°15.4′W). Schist 10–20 m from contact. Specimens span 10 m on hillside outcrop. Treatment: N–485°C, N–100 mT, 20–90 mT, 10–60 mT, 40–80 mT, 15–50 mT, N–60 mT.

Site C18 (45°2.5′N, 117°13.5′W). Schist 20 m from contact. Specimens span 5 m on hillside outcrop. Treatment: N–

547°C, N–563°C, N–90 mT, N–100 mT, N–70 mT, N–28 mT, N–60 mT, N–28 mT.

Site C22 (45°1.0′N, 117°13.0′W). Schist 50 m from contact with pluton. Specimens span 5 m in road cut. Treatment: N–540°C, N–28 mT, N–20 mT, 5–15 mT, 5–70 mT, N–20 mT, N–100 mT.

Acknowledgments. We would like to thank W. R. Dickinson, D. S. Izmirian, J. R. Magill, M. O. McWilliams, and G. W. Walker for their help. This research was supported by National Science Foundation grant EAR76-00120.

REFERENCES

Armstrong, R. L., W. H. Taubeneck, and P. O. Hales, Rb-Sr and K-Ar geochronometry of Mesozoic granitic rocks and their Sr isotopic composition, Oregon, Washington, and Idaho, *Geol. Soc. Amer. Bull., 88,* 397–411, 1977.

Beck, M. E., Discordant paleomagnetic pole positions as evidence of regional shear in the western cordillera of North America, *Amer. J. Sci., 276,* 694–712, 1976.

Beck, M. E., and C. D. Burr, Paleomagnetism and tectonic significance of the Goble volcanic series, southwestern Washington, *Geology, 7,* 175–179, 1979.

Beck, M. E., D. C. Engebretson, G. S. Gromme, E. M. Taylor, and J. N. Whitney, Paleomagnetism of the middle Tertiary Clarno Formation, northcentral Oregon: Constraint on models for tectonic rotation, *Eos Trans. AGU, 59,* 1058, 1978.

Blake, M.C., Jr., and D. L. Jones, Tectonics of the Yolla Bolly junction and its significance to the plate tectonic history of northern California, *Geol. Soc. Amer. Abstr. Program, 9,* 391, 1977.

Brown, C. E., and T. P. Thayer, Geologic map of the Canyon City quadrangle, northeastern Oregon, *Map I-447,* U.S. Geol. Surv., Reston, Va., 1966.

Carey, S. W., The tectonic approach to continental drift, in *Continental Drift Symposium,* pp. 177–358, Geology Department, University of Tasmania, Hobart, 1958.

Cox, A., Remanent magnetization of lower to middle Eocene basalt flows from Oregon, *Nature, 179,* 685–686, 1957.

Cox, A., and J. Magill, Tectonic rotation of the Oregon Coast Range, *Eos Trans. AGU, 58,* 1126, 1977.

Davis, G. A., J. W. H. Monger, and B. C. Burchfiel, Mesozoic construction of the Cordilleran "collage" central British Columbia to central California, in *Mesozoic Paleogeography of the Western United States, Pacific Coast Paleogeography Symposium,* vol. 2, edited by D. G. Howell and K. A. McDougall, pp. 1–32, Pacific Section, Soc. of Econ. Paleontol. and Mineral., Los Angeles, Calif., 1978.

Dickinson, W. R., Mesozoic forearc basin in central Oregon, *Geology, 7,* 166–170, 1979.

Dickinson, W. R., and T. P. Thayer, Paleogeographic and paleotectonic implications of Mesozoic stratigraphy and structure in the John Day inlier of central Oregon, in *Mesozoic Paleogeography of the Western United States, Pacific Coast Paleogeography Symposium,* vol. 2, edited by D. G. Howell and K. A. McDougal, pp. 147–162, Pacific Section, Soc. of Econ. Paleontol. and Mineral., Los Angeles, Calif., 1978.

Fisher, R. V., Early Tertiary deformation in north-central Oregon, *Amer. Ass. Petrol. Geol. Bull., 51,* 111–123, 1967.

Greenhaus, M. R., and A. Cox, Paleomagnetism of the Morro Rock–Islay Hill complex as evidence for crustal block rotations in central coastal California, *J. Geophys. Res., 84,* 2393–2400, 1979.

Hamilton, W., Mesozoic California and the underflow of Pacific mantle, *Geol. Soc. Amer. Bull., 80,* 2409–2430, 1969.

Hamilton, W., Mesozoic tectonics of the western United States, in *Mesozoic Paleogeography of the Western United States, Pacific Coast Paleogeography Symposium,* vol. 2, edited by D. G. Howell and K. A. McDougall, pp. 33–72, Pacific Section, Soc. of Econ. Paleontol. and Mineral., Los Angeles, Calif., 1978.

Irving, E., Paleopoles and paleolatitudes of North America and speculations about displaced terranes, *Can. J. Earth Sci., 16,* 669–694, 1979.

Jones, D. L., N. J. Silberling, and J. W. Hillhouse, Microplate tectonics of Alaska—Significance for the Mesozoic history of the Pacific coast of North America, in *Mesozoic Paleogeography of the Western United States, Pacific Coast Paleogeography Symposium,* vol. 2, edited by D. G. Howell and K. A. McDougall, pp. 71–74, Pacific Section, Soc. of Econ. Paleontol. and Mineral., Los Angeles, Calif., 1978.

Kamerling, M. J., and B. P. Luyendyk, Tectonic rotation of the Santa Monica Mountains region, western Transverse Ranges, California, suggested by paleomagnetic vectors, *Geol. Soc. Amer. Bull., Part I, 90,* 331–337, 1979.

Kawai, N., K. Hirooka, and T. Nakajima, Paleomagnetic and potassium-argon age informations supporting Cretaceous-Tertiary hypothetic bend of the main island Japan. *Palaeogeogr. Palaeoclimatol. Palaeoecol., 6,* 277–282, 1969.

Larson, R. L., and T. W. C. Hilde, A revised time scale of magnetic reversals for the Early Cretaceous and Late Jurrassic, *J. Geophys. Res., 80,* 2586–2594, 1975.

Plumley, P. W., and M. E. Beck, Tectonic rotation of Oligocene intrusive rock in the Coast Range of Oregon: A constant rate of rotation for the period 50–30 m.y. B.P., *Eos Trans. AGU, 58,* 1126, 1977.

Simpson, R. W., and A. Cox, Paleomagnetic evidence for tectonic rotation of the Oregon Coast Range, *Geology, 5,* 585–589, 1977.

Taubeneck, W. H., Geology of the Elkhorn Mountains, northeastern Oregon: Bald Mountain batholith, *Geol. Soc. Amer. Bull., 68,* 181–238, 1957.

Taubeneck, W. H., Major plutons of the Wallowa batholith, northeastern Oregon, *Geol. Soc. Amer. Abstr. Programs, 76,* 295–296, 1964*a*.

Taubeneck, W. H., Cornucopia stock, Wallowa Mountains, northeastern Oregon: Field relationships, *Geol. Soc. Amer. Bull., 75,* 1093–1116, 1964*b*.

van Hinte, J. E., A Jurassic time scale, *Amer. Ass. Petrol. Geol. Bull, 60,* 489–497, 1976.

Walker, D. W., Geologic map of Oregon east of the 121st meridian, *Map 1-902,* U.S. Geol. Surv., Reston, Va., 1977.

Waters, A. C., Basalt magma types and their tectonic associations: Pacific Northwest of the United States, in *The Crust of the Pacific Basin, Geophys. Monogr. Ser.,* vol. 6, edited by G. A. Macdonald and H. Kuno, pp. 158–170, AGU, Washington, D. C., 1962.

Watkins, N. D., Paleomagnetism of the Columbia plateaus, *J. Geophys. Res., 70,* 1379–1406, 1965.

Watkins, N. D., and A. K. Baksi, Magnetostratigraphy and oroclinal folding of the Columbia River, Steens, and Owyhee basalts in Oregon, Washington, and Idaho, *Amer. J. Sci., 274,* 148–189, 1974.

Wilkinson, W. D., and K. T. Oles, Stratigraphy and paleoenvironments of Cretaceous rocks, Mitchell Quadrangle, Oregon, *Amer. Ass. Petrol. Geol. Bull., 52,* 129–161, 1968.

Wise, D. U., An outrageous hypothesis for the tectonic pattern of the North American Cordillera, *Geol. Soc. Amer. Bull., 74,* 357–362, 1963.

Zijderveld, J. D. A., A. C. demagnetization of rocks: Analysis of results, in *Methods in Paleomagnetis, Developments in Solid Earth Geophysics,* vol. 3, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 254–286, Elsevier, Amsterdam, 1967.

(Received July 16, 1979;
accepted September 25, 1979.)